

Corporate Action Margin Impact Simulator for rToken Traders

A Project Specification for the Bitget Base Camp Hackathon S2

Track: AI Trading Desk (Track 3)

Theme: Open Theme

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1. The Idea Behind What We're Building

Bitget's rTokens are designed to reflect corporate actions automatically. When a stock splits, token quantity and unit price adjust proportionally. When dividends are paid, they are converted to USDT and credited. For spot holders, this process is seamless.

But for traders using rTokens as leveraged margin in Bitget's Unified Trading Account (UTA), the downstream consequences are invisible until they hit. Eligible rTokens contribute collateral-adjusted value to a trader's margin pool, with collateral ratios reaching up to 95%. When a corporate action occurs, the mechanical adjustment changes token quantity and unit

price—but the margin consequence (collateral value, liquidation threshold, margin ratio) is not pre-computed anywhere in Bitget's documentation or interface.

The gap is real. Bitget provides the formulas: Adjusted Equity = Σ (collateral-ratio-adjusted value), and Account Leverage = Total Position Value $\div$ Adjusted Equity. But no tool bridges the gap between "here's what the corporate action does to your tokens" and "here's what it does to your leveraged position."

Our tool fills that gap. It ingests corporate action events, takes the user's specific portfolio state, applies Bitget's UTA margin formulas, and generates a plain-English report showing: new collateral value, new margin ratio, whether liquidation risk increases, and a recommended action—before the event hits.

The One-Sentence Pitch:

A research workbench that tells leveraged rToken traders what a corporate action will do to their margin—before it happens, not after.

2. Target User

Retail rToken traders using 2x–5x leverage in UTA, holding rTokens as collateral, who cannot manually cross-reference corporate action announcements, collateral ratio tables, and maintenance margin formulas fast enough to act before the event.

This segment is specific and underserved. It is not "all traders."

3. LLM's Role

The LLM performs three non-trivial jobs:

1. Extraction: Parse corporate action events from Alpaca into structured data.
2. Reasoning: Apply UTA margin formulas to a specific portfolio state and event.
3. Generation: Explain results in plain English with a recommended action.

The margin math is deterministic. The LLM explains it; it does not invent it.

4. Complete Build Requirements

4.1 Data Sources

Item What It's For Source

Corporate action events Forward splits, reverse splits, cash dividends, stock dividends, mergers, spin-offs Alpaca Corporate Actions API
 rToken list + collateral ratios Which rTokens are eligible as UTA margin, what their ratios are (up to 95%) Bitget public documentation
 UTA margin formulas Adjusted Equity, Account Leverage, liquidation mechanics Bitget UTA documentation
 Historical corporate actions For backtesting simulation against real past events Alpaca API + SEC EDGAR
 Live rToken prices Current market data for portfolio valuation Bitget Agent Hub market module
 Account/margin state User's actual UTA position (for live mode) Bitget Agent Hub account module

4.2 Technical Stack

Layer What It Is Source/Notes

LLM Qwen (via Bitget hackathon credits) Apply through separate Google Form; first 300 teams get 30U equivalent
 Agent framework Bitget Agent Hub (MCP Server or CLI) bgc CLI or MCP Server for Claude/Cursor
 Research Skills bitget-signal package macro-analyst, market-intel, news-briefing, sentiment-analyst, technical-analysis. No API key needed
 Corporate action ingestion Python/JS wrapper for Alpaca API Free tier: 200 calls/min
 Margin simulation engine Custom code Deterministic math applying UTA formulas
 Demo UI Streamlit / Gradio / simple web app Must be accessible via link for judges
 Hosting Vercel / Railway / Hugging Face Spaces Free tier sufficient for demo

4.3 Bitget-Specific Tools

Tool Purpose Access

Bitget Agent Hub Trading tools platform for AI developers GitHub
 MCP Server One-line config for Claude Desktop/Cursor/Windsurf Part of Agent Hub
 CLI (bgc) Terminal trading tool for Claude Code/Codex Part of Agent Hub
 Agentic Account Dedicated sub-account with isolated funds, no withdrawals OAuth connection, no manual API key
 bitget-signal Skills Pre-packaged research logic No account or API key needed
 Bitget Playbook Quant strategy platform for backtesting Bitget account login

4.4 Submission Deliverables

Deliverable Requirement Where

Project Description 6 parts: Thesis, Target User, Validation, Progress, Deliverables, AI Trading take Google Form field
 Role of the LLM What model does, which models used, Qwen usage if applicable Separate Google Form field
 Submission Materials Link Demo + code + any logs Single Google Form field
 X Promotional Post Must include #BitgetHackathon + @Bitget_AI, introduce product Separate Google Form field

Accessible Demo Working, clickable Hosted link
Complete Research Task Demo Question → actionable insight flow Demo or video
Track → Sub-theme Select AI Trading Desk → Open Theme Google Form field
University Name If applicable Optional field
Apply for Demo Day Check box Optional
Apply for K3 Token Subsidy Check box Optional

4.5 Build Timeline (19 Days Remaining)

Days Focus Deliverable

1-2 Finalize thesis, user persona, validation plan Written project description draft
3-5 Build corporate action ingestion (Alpaca API) Script pulling events for 5 rTokens
6-9 Build margin simulation engine Code applying UTA formulas to portfolio + event
10-12 Build LLM layer (Qwen + Agent Hub Skills) Report generation from simulation output
13-14 Build demo UI (Streamlit) Clickable demo
15-16 Run validation (5 portfolios, 3 events) Results table
17 Write submission, record video, post on X Complete submission
18 Submit Google Form Submission complete
19 Buffer —

5. What You Do NOT Need

- A live trading Agent (that is the Agentic Trading track)
- 2 weeks of paper trading logs (that is the Agentic Trading track)
- A Sharpe ratio (that is the Alpha Factory track)
- A team (solo is allowed)
- Expensive AI models (Qwen credits cover it)
- Paid data subscriptions (Alpaca free tier + Bitget public data suffice)
- Real users (validation plan with simulated portfolios is accepted)

6. Access Audit: What's Available Now

6.1 Immediately Accessible (No Cost, No Permission)

Item How to Access Notes

Alpaca Corporate Actions API <https://data.alpaca.markets/v1/corporate-actions> Free with API key. Supports forward_split, reverse_split, cash_dividend, stock_dividend, spin_off, cash_merger, stock_merger

Alpaca Market Data API <https://docs.alpaca.markets/> Free tier: 200 req/min, real-time IEX feed, historical data since 2016

Bitget rToken Documentation Bitget Academy rToken FAQ Covers mechanics, collateral ratios, corporate action handling

Bitget UTA Documentation Bitget support articles Margin formulas, collateral ratios (up to 95%)

Bitget Agent Hub GitHub https://github.com/BitgetLimited/agent_hub MCP Server, CLI, 89 operations across 14 intent verbs
bitget-signal Skills Part of Agent Hub 5 research skills: macro-analyst, market-intel, news-briefing, sentiment-analyst, technical-analysis
Bitget Playbook Bitget account login Strategy backtesting platform

6.2 Requires Action (Apply/Register)

Item Action Required Where

Qwen Token Credits (30U) Fill separate Google Form; pass Bitget KYC Qwen application form

Bitget KYC Complete identity verification Bitget account

Agentic Account OAuth authorization (no manual API key) Via Agent Hub prompt

K3 Token Subsidy (30U) Select "Apply for K3 Token Subsidy" in submission form Google Form

6.3 With Limitations

Item Limitation Workaround

Alpaca Corporate Actions timing "No guarantees on creation time. May not be available immediately after announced" Use historical events for demo; state this limitation openly

Bitget Agent Hub account module Requires Agentic account authorization Complete OAuth flow; or use read-only mode for demo

Live rToken prices Market data available without API key Use Agent Hub market module

Demo hosting Free tier limits on Vercel/Railway Sufficient for hackathon demo

7. Key Links

Resource URL

Alpaca Corporate Actions API <https://data.alpaca.markets/v1/corporate-actions>

Alpaca Market Data Docs <https://docs.alpaca.markets/>

Bitget Agent Hub GitHub https://github.com/BitgetLimited/agent_hub

Bitget rToken FAQ Bitget Academy

Bitget UTA Collateral Guide Bitget Academy

Bitget Playbook <https://www.bitget.com/activity/ai-get-agent/playbook>

Hackathon Submission Form <https://forms.gle/GyWZCMCPocgJdJon6>

Official Telegram https://t.me/+o1tYqQ_IXxIIYjgy

8. Risk Assessment

8.1 What's Solid

- The gap is real: no tool pre-computes corporate action margin impact.
- Data sources exist: Alpaca API (free), Bitget Agent Hub (available).

- Track fits: AI Trading Desk + Open Theme is 100% subjective scoring.
- Timeline is tight but feasible with scoped demo.

8.2 What's Risky

- Alpaca corporate action timing delay—use historical events for demo.
- Agentic account authorization may take time—start KYC immediately.
- 19 days is short—scope to 5 rTokens, 3 event types, 1 user persona.

8.3 What's Unresolved

- Whether Bitget's UTA interface already shows projected margin impact somewhere not found in docs.
- Whether other teams are building similar tools (unknowable).